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Seaweed-derived biochar with multiple active sites as a heterogeneous catalyst for converting macroalgae into acid-free biooil containing abundant ester and sugar substances

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ABSTRACT

A series of environmentally friendly and low-cost seaweed-derived biochar catalysts (SDBCs) were prepared. Results indicated that metals modified SDBCs at high temperature displayed highly developed porous structures and more aromatic rings with no less than 6 benzene rings along with defects and heteroatoms. The inorganic minerals were mostly present in the form of NaCl and KCl in the seaweed-derived biochars. Subsequently, the performance of SDBCs was examined through ex-situ catalytic pyrolysis of *Enteromorpha clathrate*. Results demonstrated that biooils from catalytic pyrolysis exhibited higher $(H/C)_{eff}$ and HHVs in comparison to the control. Interestingly, the carboxylic acids were decreased significantly after using SDBCs. The mechanism is that a significant amount of hydroxyl groups existed in biochar could react with carboxylic group to form FAMEs. Additionally, the metals Cu, Ce and Fe loaded on the BC-800 catalyst had particular effects on the C–C bond and -C = O- units in the pyrolysis volatiles. It was suggested that Cu-BC-800 catalyst was a promising metal modified biochar catalyst for producing acid-free biooils containing abundant esters and sugars.

1. Introduction

Pyrolysis technology provides broad prospects for the harmless treatment of biomass waste. However, due to the low quality of the reaction products, such as high oxygen/acid contents in biooil, low calorific value of non-condensable gas, it is difficult to be widely utilized in the industrial scale [1–3,52]. At present, many researchers have been conducting researches to improve the process conditions to promote the quality of pyrolysis products, among which catalytic pyrolysis is considered to be one of the most promising methods. Recently, several catalysts have been examined such as; inorganic minerals (K, Ca, Na, Mg, etc.) [4,5], metal oxides (Al_2O_3, SiO_2, MgO, CaO, Fe_2O_3, etc.) [6,7], and zeolite catalysts (ZSM-5, HY, Hβ, MCM-41, etc.) [8-11]. Carbon-based materials, such as biochar obtained from biomass pyrolysis or gasification, have received extensive attention as a catalyst or catalyst support. Biochar is an inevitable by-product of biomass pyrolysis and gasification; whose utilization covered the fields of soil remediation, waste water treatment, greenhouse gas reduction, biofuel upgrading, and energy storage equipment (e.g., supercapacitors, lithium-ion batteries, and fuel cells) nowadays [12]. Studies have shown that biochar could affect the cracking reforming reactions of the pyrolysis volatiles, and remove the tars during biomass gasification [13-15].

Compared with the other catalysts, char-based catalysts have several advantages. Firstly, biochar itself contains small amounts of inorganic metal minerals. Those inorganic metal minerals have a particular effect on the pyrolysis pathway of biomass [16]. Alkali and alkaline earth metals (K, Ca, Na, Mg) are the most abundant metal elements in biomass. The metal ion is in conjunction with the hydroxyl and aldehyde groups forming a metal ion-water complex, which can effectively induce the dehydration reaction in the pyrolysis [17]. Secondly, biochar has abundant heterogeneous functional groups on its surface, including oxygen species (carboxylic groups, phenolic hydroxyl groups, carbonyl groups, and ester groups) or/and nitrogen-containing species (pyridine-N, pyrrole-N, quaternary-N, etc.) [18,51]. Those chemical functional groups act not only as absorbents but also catalysts during pyrolysis [19]. Additionally, the biochar physicochemical properties can be

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facilely tailored compared with other carbon materials such as carbon black and activated carbon from coal coke [20].

Most of the researchers have been focused on the interaction between coal or terrestrial biochar and volatiles produced from pyrolysis or gasification procedure. Shen Y. et al. studied the catalytic performance of conversion of tar over rice husk char-supported nickel–iron catalysts [19,21]. In their experiment, the conversion efficiency of heavy tar to light components could be as high as 92.3%, and the conversion efficiency of CO_2 to CO was promoted, improving the quality of gaseous products. Chen W. et al. explored the co-pyrolysis and deoxygenation reaction of bamboo wastes and microalgae over pyrolysis biochar from bamboo wastes [22]. The results showed that biochar-based catalysts had a greater effect on the chemical components of co-pyrolysis liquid products, which could promote the cracking process of long-chain fatty acids and oxygenated substances. They also investigated the catalytic fast pyrolysis of bamboo wastes for phenols products over N-doped biochar catalysts [23]. Results showed that N-doped biochar catalyst prominently promoted the generation of phenols, which reached up to 82%. Zhang Y. et al. investigated the selective monophenol production from cellulose pyrolysis over corn stover activated carbons through microwave-induced pyrolysis [24]. This kind of catalyst had an excellent catalytic performance to selectively produce phenols.

However, there were few studies reporting the catalytic performance of seaweed biochar-based catalysts. Taghavi S. et al. used Venice lagoon brown marine algae to prepare algal biochar catalyst for producing H_2-rich gas, and valuable biochemical [25]. Besides, they compared the algal biochar and Ni/SBA-15 in terms of their catalytic performance. Both algal biochar and Ni/SBA-15 could upgrade and refine biooil to more valuable chemicals and enhance hydrogen-rich gas production. The biochar catalyst exhibited excellent selectivity toward phenols, while Ni/SBA-15 showed good performance for promoting H_2 production. Algae biomass comprise high alkali/alkaline earth metal, ash, and nitrogen [26]. For example, *Enteromorpha clathrate (E. clathrate),* as a typical seaweed biomass, has an ash content of over 20–28 wt% [27-29]. Algae contain macromolecules such as protein, containing high content of nitrogen (4–7 wt%) [27,28,30]. In addition, *E. clathrate* contains a high amount of K and Na due to its specific marine growth environment. The contents of K and Na in *E. clathrate* were 5.27 wt% and 9.61 wt%, respectively in our previous study [31]. Therefore, these characteristics of seaweed provide a unique advantage for further preparation of highly efficient biochar catalysts or catalyst supports compared to the other terrestrial biomass.

This work aims to prepare a series of environmentally friendly, easily obtainable and low-cost seaweed-derived biochar catalysts (SDBCs) from seaweed *E. clathrate*. SDBCs were morphologically and chemically characterized through scanning electron microscopy (SEM), N_2-adsorption/desorption, Fourier Transform infrared spectroscopy (FTIR), Raman spectroscopy (Raman) and X-ray diffraction (XRD). The catalytic pyrolysis process was performed using SDBCs to upgrade and refine the pyrolysis volatiles from green macroalgae, *E. clathrate*. Additionally, the catalytic mechanism over the synthesized SDBCs was studied.

2. Materials and methods

2.1. Materials

Seaweed employed for preparing biochar catalysts in this study was *E. clathrate*, which was collected from Zhanjiang coast of South China Sea, Guangdong Province, China. *E. clathrate* also was used as the raw material for pyrolysis in the measurement of catalytic performance using SDBCs. The proximate and ultimate analyses were previously ascribed in our work [29]. The relative contents of the main biochemical compounds (polysaccharides, proteins and lipids) in *E. clathrate* are 36.84%, 22.10%, and 1.18%, respectively [32].

2.2. Preparation of SDBCs

Seaweed-derived biochar catalyst was prepared via fast pyrolysis of *E. clathrate*. 5 g of dried raw materials were fed into a stainless reactor when the temperature was stable at target temperatures. Three temperatures, 600 °C, 700 °C, and 800 °C were chosen as the target in this study. The carrier gas was nitrogen, which could create an oxygen-free condition. Its flow rate was set as 0.2 L/min during the pyrolysis for the reaction time of 1 h. The apparatus was shown in our previous literature [29,33]. The resulting solid products were collected in sealed bags and were marked as BC-600, BC-700 and BC-800.

All metals were loaded to modify BC-800 by incipient wetness impregnation method. 1 mmol $Cu(NO_3)_2{\cdot}3H_2O$, $Ce(NO_3)_3{\cdot}6H_2O$ or $Fe(NO_3)_3{\cdot}9H_2O$ were separately dissolved in 60 mL of deionized water and then mixed with 1 g of prepared BC-800. The resulting solutions were stirred for 24 h at 80 °C. After being dried by evaporation, the samples were separately calcined at 800 °C for 3 h under the N_2. The resulting products were collected in sealed bags and marked as M−BC−800 (M: Cu, Ce, Fe).

2.3. Characterizations of SDBCs

Simultaneous thermal analysis analyzer, which was STA 449 F3 instrument from NETZSCH-Gerätebau GmbH, was employed to measure the thermal stability and thermal decomposition characterizations of SDBCs. In a typical analysis, 2–5 mg of samples was fed into an alumina crucible and pyrolyzed in a high purity nitrogen gas at a flow rate of 100 mL/min. The reaction temperature was raised from room temperature to 1000 °C at a constant heating rate of 10 °C/min. The morphology of the SDBCs was investigated by scanning electron microscopy (SEM) using a Hitachi microscope (S-4700, Hitachi, Japan) operated at an acceleration voltage of 10 kV. The element distribution was determined by energy dispersive spectrometer (EDS). The porous characteristics of the SDBCs was measured with N_2-adsorption/desorption at a liquid nitrogen temperature of 77 K using a Micromeritics micropore analyzer (ASAP2020, USA). The Brunauer-Emmett-Teller (BET) method was employed to determine mesopore-enclosed surfaces of the biochar catalysts. The classical model (BJH) was used in the characterization of porous biochar catalysts based on the adsorption isotherm [34]. The FTIR measurements were performed on a Nicolet Nexus 470 1.2 microscopic infrared spectrometer. The scan region ranged from 4000 cm^{-1} to 400 cm^{-1} and the FTIR spectra were recorded by the number of scanning of 32 at a resolution of 2 cm^{-1}. A pulverous mixture of the SDBCs and KBr with mass ratio of 1:100 was employed. The Raman measurement was conducted to study the microcrystalline structure using a Laser Raman Spectrometer equipped with a 532 nm laser from ThermoFisher, America. PeakFit v4.12 was employed to fit the peaks of the Raman spectra between 800 and 1800 cm^{-1}, and the peak center, full width at half maximum and integral area for each curve-fitted peak were obtained. XRD measurements were implemented to explore the crystal structure of the SDBCs, and the catalysts were radiated by Cu-Kα (λ = 1.5418 Å) within the 2θ = 10–80° range using a diffractometer named Bruker AXS GmbH from Germany.

2.4. Catalytic pyrolysis of macroalgae

The *ex-situ* catalytic pyrolysis experiments were performed in a fixed bed quartz tube reactor. The schematic diagram is shown in Fig. 1. Nitrogen gas was used as the carrier gas. As soon as the temperature of the nitrogen gas preheating equipment reached around 550 °C, the nitrogen gas was fed at a flow rate of 800 mL/min to keep an oxygen-free atmosphere in the reactor before the pyrolysis of *E. clathrate*. After the reactor was heated to 550 °C, the prepared *E. clathrate* sample, stored in the sample storage connected with the fixed bed reactor, was fed into the reactor after the nitrogen gas flow rate was adjusted to 200 mL/min. The condensable volatiles were collected in the condenser as the biooil. The

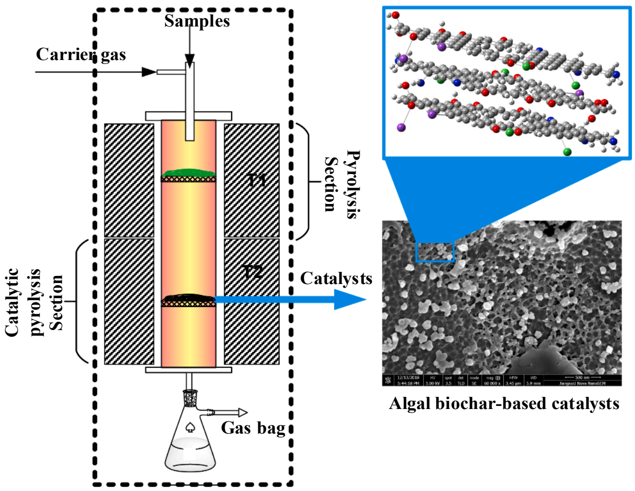

Fig. 1. Schematic diagram of the catalytic pyrolysis of macroalgae.

solid residues were collected in the fixed bed reactor as the biochar. The weight of biooil and biochar were directly measured. The weight of gas products was calculated from the results of gas chromatograph. The yields of solid, liquid and gaseous products from the catalytic pyrolysis in the experiments were then calculated using Eqs. (1), (2) and (3) below:

$$\text{Biooil yield (wt. \%)} \;=) = W_{\text{liquid}}/W_{\text{tol.}} \times 100\% \tag{1}$$

$$\text{Biochar yield (wt. \%)} \;=) = W_{\text{solid}}/W_{\text{tol.}} \times 100\% \tag{2}$$

$$\text{Non - condensable gas yield (wt. \%)} \;=) = W_{\text{gas}}/W_{\text{tol.}} \times 100\% \tag{3}$$

where W_{liquid}, W_{solid}, W_{gas} and $W_{tol.}$ represent the weight of biooil, biochar, non-condensable gas and the dry weight of injected biomass (g), respectively.

2.5. Products analysis

2.5.1. Element analysis of biooils

Ultimate analysis (C, H, N, S, and O) of the liquid products were performed on vario EL III, and the higher heating values (HHVs) of liquid products were calculated by the modified Dulong formula as follow [35,36];

$$HHV(MJ/\text{kg}) = 0.3578 \times [\%C] + 1.1357 \times [\%H] - 0.0845 \times [\%O] + 0.0594 \times [\%N] + 0.119 \times [\%S] \tag{4}$$

The effective hydrogen to carbon ratio ($(H/C)_{eff}$) could be a useful comparative parameter for the upgradation of biooils in existing refineries. The higher $(H/C)_{eff}$ was, the more efficient conversion was in refineries. The ratio that feedstocks containing heteroatoms was defined by Eq. (5);

$$(H/C)_{eff} = (H - 2O - 3N - 2S)/C \tag{5}$$

where H, C, O, N and S were the molar ratio of the corresponding elements on dry basis.

2.5.2. GC–MS analysis of biooils

The chemical compounds of *E. clathrate* catalytic pyrolysis liquid products were detected through a gas chromatograph (Agilent Technologies, 7890A) coupled with a mass spectrometer (Agilent Technologies, 5975C) with a triple-axis detector. The GC system was equipped with an Agilent HP-5 UI column (30 m (length) × 250 μm (diameter) × 0.25 μm (thickness)). High-purity helium was used as the carrier gas at a flow rate of 3 $mL\cdot min^{-1}$. After the temperature of the GC injector was up to 300 °C, the mixed solution of pyrolysis oils and anhydrous alcohol was injected into the GC–MS system of injection volume 1 μL. The GC oven was heated from an initial temperature of 40 °C to 300 °C at a ramping heating rate of 10 °C/min. It was maintained at 300 °C for 5 min.

2.5.3. GC-FID/TCD analysis of non-condensable gas

The non-condensable gas was analyzed by the gas chromatograph equipped with a flame ionization detector and thermal conductivity detector (GC-FID/TCD). A KB-Al_2O_3/Na_2SO_4 capillary column (50 m × 0.53 mm × 20.00 nm) was employed to separate the light hydrocarbon gases (CH_4, C_2H_6, C_2H_4, C_3H_8, and C_3H_6). A TDX-01 packed column (2m × 3mm) was used to analyze H_2, N_2, CO, CO_2. High-purity argon (99.999%) was employed as the carrier gas. Standard gas mixture was used for regular calibration, which consists of H_2 (10.070%), N_2 (70.878%), CO (5.000%), CO_2 (5.020%), CH_4 (4.980%), C_2H_6 (0.993%), C_2H_4 (1.050%), C_3H_8 (0.999%), and C_3H_6 (1.010%). Each sample was detected three times to calculate their average.

3. Results and discussion

3.1. Characterizations of SDBCs

3.1.1. TGA analysis

Fig. 2 represents the TGA curves of SDBCs. For BC-600, BC-700 and BC-800, there was a gentle weight loss of the samples before reaching the temperatures of 600, 700 and 800 °C, respectively. After those temperatures, the catalysts would decompose sharply from the TG curves in Fig. 2(a). The inorganic minerals decomposed after the temperature of 800 °C. The metal modified biochar catalysts (Cu-BC-800, Ce-BC-800 and Fe-BC-800) showed poorer thermal stability than BC-800 due to the higher weight loss percentages from the TG curves in Fig. 2 (b). There was also a gentle weight loss of the sample before reaching the temperature of 800 °C. While, after the temperature exceeded 800 °C, the Cu-BC-800, Ce-BC-800 and Fe-BC-800 catalysts decomposed significantly.

The catalytic performances of all the catalysts would be tested under the condition of 550 °C. The weight loss percentage of BC-800 at the temperature of 550 °C was the highest (7.40%). TG curves of BC-700 and BC-800 before 550 °C coincided. Therefore, their weight loss percentages at 550 °C was identical (4.48%). The weight loss percentages of Cu-BC-800, Ce-BC-800 and Fe-BC-800 were higher than BC-800 at 550 °C, which were 6.14%, 6.14% and 11.67%, respectively. TG curves of Cu-BC-800 and Ce-BC-800 before 550 °C coincided.

3.1.2. SEM

SEM was used to characterize the surface morphology of SDBCs. As shown in Fig. 3, all the SDBCs exhibited irregular porous structures. These pores were formed by the rapid release of volatiles from the algae biomass particles during the devolatilization process. As the preparation temperature increased, the morphology also showed a definite difference. Furthermore, the surface morphology of the BC-800 catalyst after metal modified showed a significant difference from BC-800. Besides, some small and bright crystals can be seen on the surface of all the SDBCs from Fig. 3.

The EDS results of Cu-BC-800, Ce-BC-800 and Fe-BC-800 indicated that the most-abundant metallic elements existed in the catalysts were K and Na. In addition, the content of Cl was high, which was owing to the growth environment of seaweed. Amongst, potassium could act as one kind of multiple active sites, which effectively promoted the dehydration, demethoxylation and secondary reactions in the pyrolysis [17]. Hwang H. et al. concluded that water content and small molecules in bio-oil increased due to the dehydration, demethoxylation and secondary reactions promoted by potassium. The loaded metals were also detected from the EDS results, which meant that Cu, Ce and Fe had been

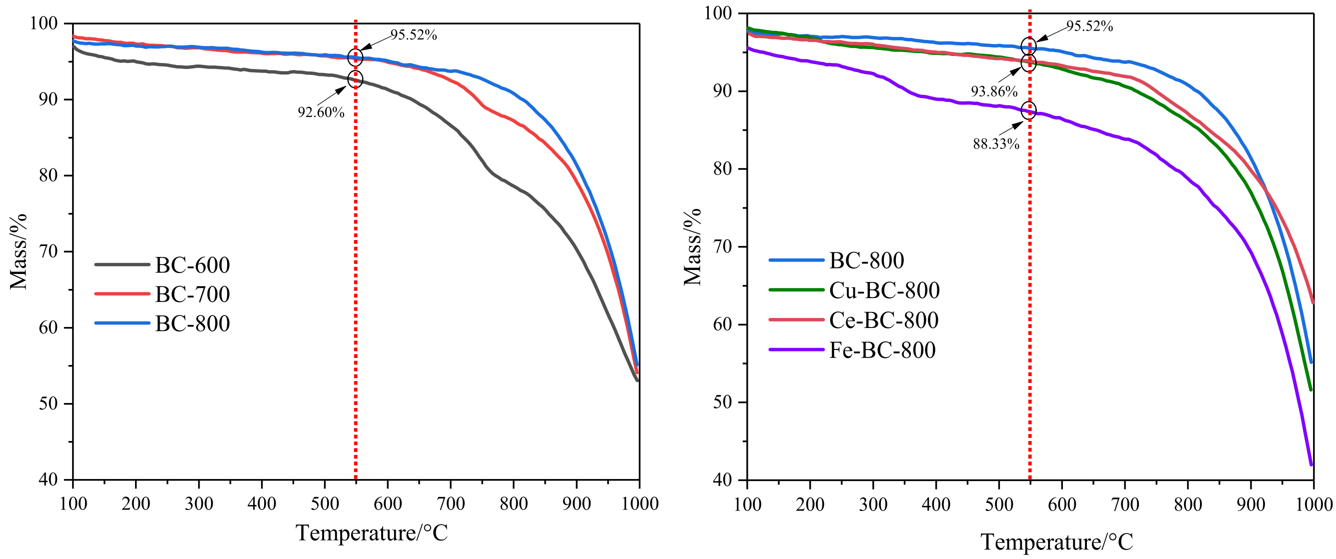

Fig. 2. Thermogravimetric analysis of SDBCs, (a) BC-600, BC-700, BC-800; (b) Cu-BC-800, Ce-BC-800, Fe-BC-800.

loaded on the BC-800 supports successfully. The weight percentages of Cu, Ce and Fe were 1.98%, 3.81% and 5.49%, respectively.

3.1.3. BET analysis

The N_2 adsorption–desorption isotherms and the pore size distribution of SDBCs are shown in Fig. 4. All the adsorption isotherms were ascribed to Type II. The adsorption capacity increased under low relative pressure (P/P_o), which indicated that there existed some micropores in the SDBCs. The adsorption isotherms at low P/P_o range of Cu/Ce/Fe-BC-800 in Fig. 4(b) increased more rapidly than that of BC-800. This indicated that the metal loaded BC-800 catalyst possessed more microporous structures than BC-800. The capillary condensation appeared more clearly as the N_2 adsorption capacity increased when the P/P_o increased up to 0.4. The phenomenon was mainly caused by the condensation of the gas phase into the liquid phase, which entered into the microporous or mesoporous structure in biochar catalysts. When the P/Po was in the range of 0.5 and 0.9, there were obvious hysteresis loops in all samples, and the adsorption capacity increased significantly with the temperature increasing. This indicated that there existed mesoporous and macroporous structures in the biochar catalysts. In addition, the adsorption capacity increased significantly after the Cu, Ce and Fe were loaded on the BC-800 catalyst. This indicated that the pore structure of the catalysts modified with metals (Cu, Ce and Fe) were more developed.

The surface area of the SDBCs is shown in Table 1. It was observed that the surface area of SDBCs increased with the increase of the preparation temperature, and the surface areas of the catalysts modified with metals (Cu, Ce and Fe) were higher. According to the pore size distribution, the SDBCs were mesoporous materials. In detail, with the increase of preparation temperature of SDBCs from 600 to 800 °C, the surface area increased from 8.38 to 22.853 m^2/g, the pore volume raised from 0.0251 to 0.0505 cm^3/g, and the average pore diameter decreased from 11.993 to 8.840 nm. After modification with metal, the surface area and the pore volume increased continuously, reaching up to 99.783 m^2/g and 0.1216 cm^3/g in the case of Fe-BC-800, while the average pore diameter was reduced. The lowest average pore diameter appeared in the case of Ce-BC-800, which was 4.145 nm. Additionally, after all metals modified, the phenomenon of the increase in the surface area might be due to the carbothermal reduction in the preparation process of catalysts. During the carbothermal reduction, the carbon in biochar catalysts were expended to some extent and some small-molecular gases were formed to activate the biochar catalysts.

3.1.4. FTIR analysis

The FTIR spectra of SDBCs are shown in Fig. 5. The FTIR spectra characteristic peaks of all SDBCs were similar in the position distributions; however, the intensity of the bands was different. The band at 3200–3600 cm^{-1} was assigned to –OH group stretching [37], where SDBCs had a peak at 3434 cm^{-1}. The –OH indicated the presence of phenol or alcohol in the seaweed-derived biochar samples. The shape of the absorption bands was broad and strong for all the SDBCs except the case of Ce-BC-800. The bands at 2923 and 2857 cm^{-1} were assigned to aliphatic $–CH_2–$ stretching only in Cu-BC-800 catalysts [38], which was a weak band. Those bands were not found in the other biochar catalysts. This illustrated that there was less aliphatic $–CH_2$- structure in biochar catalysts. The band at 1638 cm^{-1} belonged to the carbonyl group (C=O) stretching in carboxylic and ester groups. The band at 1435 cm^{-1} was mainly due to the asymmetric deformation of $–CH_2–$ and the symmetric bending of the $–CH_3$ group, indicating the presence of aliphatic pendant or methylene bridges between the aromatic rings existed in the SDBCs. The strong bands at the range of 1300–1000 cm^{-1} were attributed to the phenolic deformation C–O–C (stretching) in BC-600 and BC-700 [39], which disappeared in BC-800. This indicated that the C-O-C group was eliminated after the preparation temperature reached up to 800 °C. The absorption band between 900 and 700 cm^{-1} was attributed to the deformation vibration of –CH outside the aromatic ring [40], which was not found in BC-600 and BC-700.

3.1.5. Raman analysis

The Raman absorption spectra in the range of 800–1800 cm^{-1} were curve-fitted to gain insights into the structural features of the SDBCs. As shown in Fig. 6, the Raman spectra were deconvoluted into five bands: D_1, D_2, D_3, G and G_L. The D_1 band near the peak center of 1350 cm^{-1} represented highly regularized (no less than 6 benzene rings) carbon with defects and heteroatoms [41]. The G peak near 1580 cm^{-1} mainly represented the aromatic ring quadrant vibration and the $E2g^2$ vibration of the graphite. The D_2 band between D_1 and G bands centered around 1500 cm^{-1}, which was mainly ascribed to semi-circle breathing of aromatic rings and/or typical structures in amorphous carbon, such as the small aromatic ring polymerization degree (3–5 polymerized aromatic rings system). D_3 band near 1230 cm^{-1} represented the aryl–alkyl ether. The D_2 and D_3 bands represented the reactive sites on the biochar catalysts [42]. G_L near 1700 cm^{-1} represented an absorption band of a carbonyl C=O structure in biochar [43]. The areas of D_1, D_2 and D_3 bands were mostly stronger in comparison to the G band, indicating the existence of a large number of amorphous carbons in the carbon

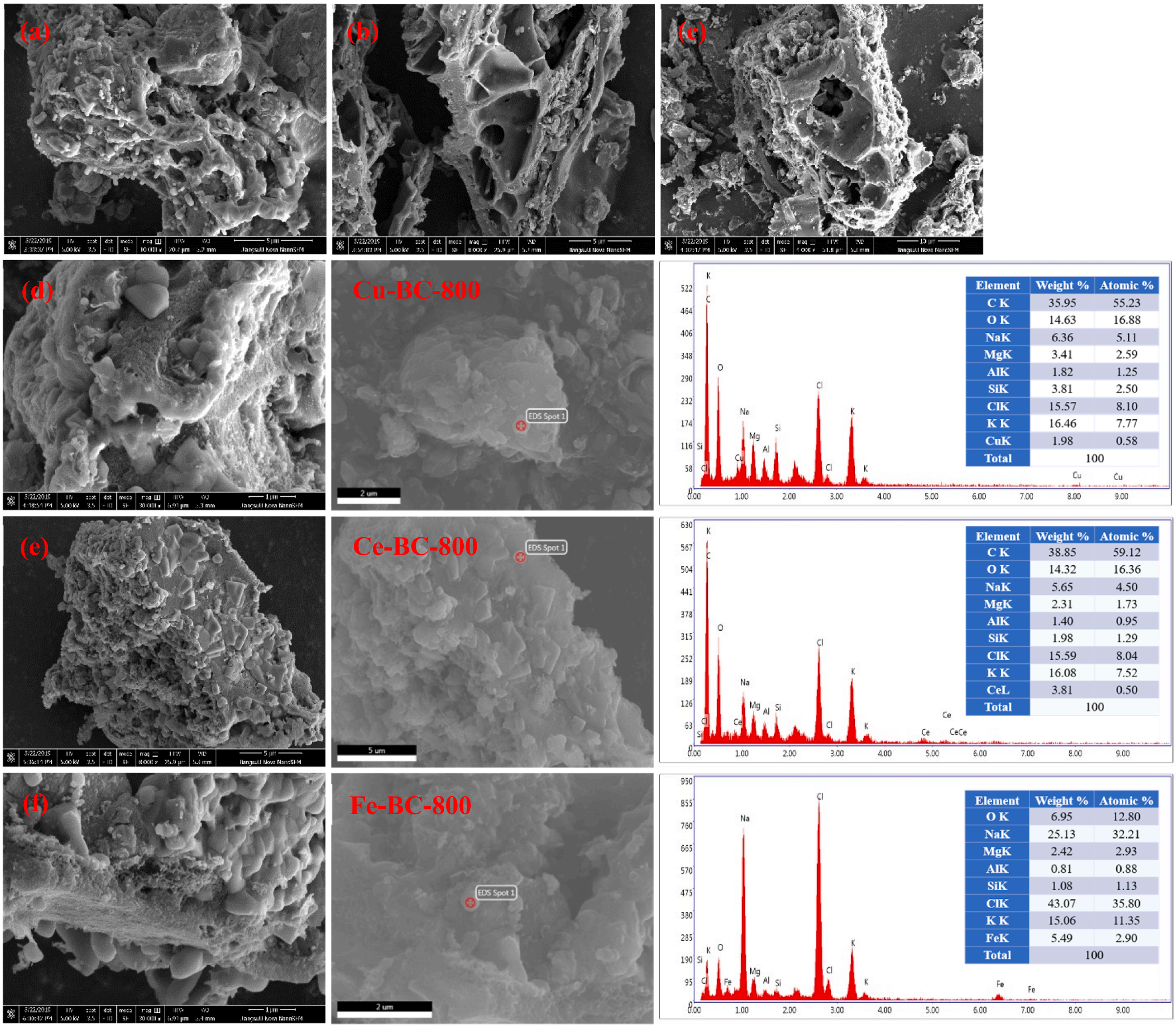

Fig. 3. Surface morphology of SDBCs from SEM-EDS (a) BC-600; (b) BC-700; (c) BC-800; (d) Cu-BC-800; (e) Ce-BC-800; (f) Fe-BC-800.

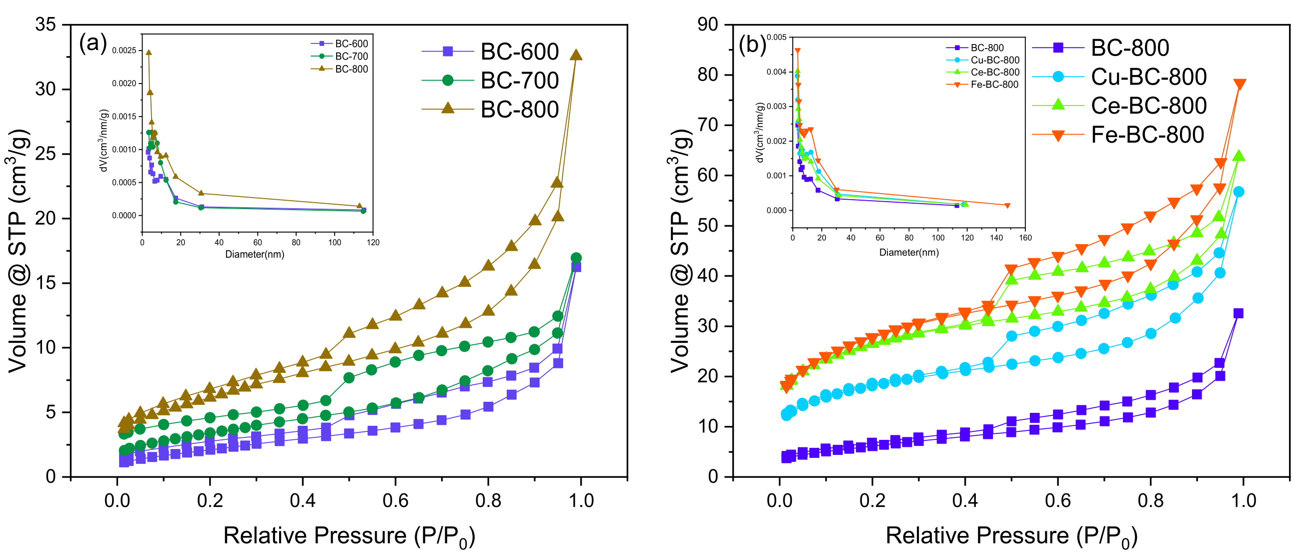

Fig. 4. N_2 adsorption–desorption isotherms and pore size distribution of (a) BC-600, BC-700, BC-800; (b) Cu-BC-800, Ce-BC-800, Fe-BC-800.

Table 1
Surface area of the SDBCs.

Parameter	Amount					
	BC-600	BC-700	BC-800	Cu-BC-800	Ce-BC-800	Fe-BC-800
BET Surface Area (m^2/g)	8.380	12.622	22.853	65.168	95.086	99.783
Pore Volume (cm^3/g)	0.0251	0.0263	0.0505	0.0880	0.0985	0.1216
Average pore diameter (nm)	11.993	8.333	8.840	5.403	4.145	4.876

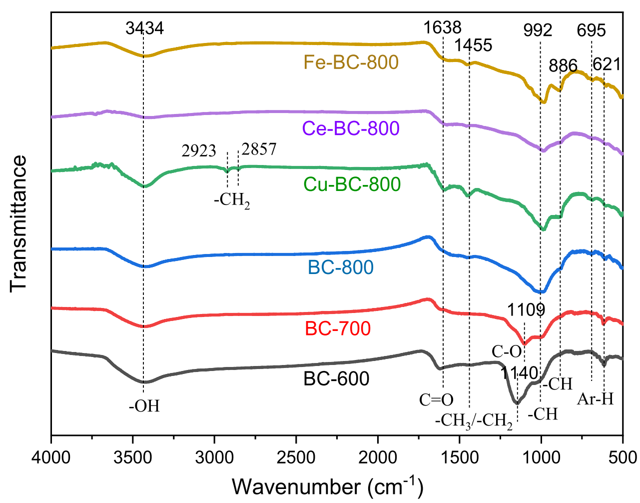

Fig. 5. FTIR analysis of SDBCs.

structure of biochar catalysts.

The band area ratios, including $I(D_1)/I(G)$, $I(D_2+D_3)/I(G)$ and $I(G)/I(All)$ were calculated to investigate the evolution of char crystalline structure (results are shown in Fig. 7). The $I(D_1)/I(G)$ ratio increased continuously from 1.525 to 1.822 with pyrolysis temperature rising from 600 °C to 800 °C, while the $I(G)/I(All)$ ratio showed the opposite trend. The results indicated an increase in the concentration of large aromatic rings containing no less than 6 benzene rings with defects and heteroatoms due to dehydrogenation of hydroaromatic [44]. The ratio of $I(D_2+D_3)/I(G)$ indicated the amorphous carbon content in biochar. The ratio of $I(D_2+D_3)/I(G)$ increased, which indicated that more amorphous carbon was formed when the pyrolysis temperature increased.

After the metals were loaded and calcinated, the $I(D_1)/I(G)$ and $I(D_2+D_3)/I(G)$ ratios of Cu-BC-800, Ce-BC-800 and Fe-BC-800 were higher than BC-800, while the $I(G)/I(All)$ ratios were lower than BC-800. This indicated that loading metals such as Cu, Ce and Fe and calcination could result in more aromatic rings with no less than 6 benzene rings along with defects and heteroatoms. Also, more amorphous carbon was formed in biochar catalysts.

3.1.6. XRD analysis

The identification of crystal phases was performed by XRD analysis. A number of crystalline and semi-crystalline phases in the SDBCs were close to the XRD detection limit, which were belonged to minerals commonly Na and K. The Na and K in SDBCs were mostly present in the form of NaCl and KCl (as shown in Fig. 8), which was consistent with the SEM-EDS analysis on the surface elements of SDBCs. According to Fig. 8 (a), the metal components in the structure were mainly K and Na, and

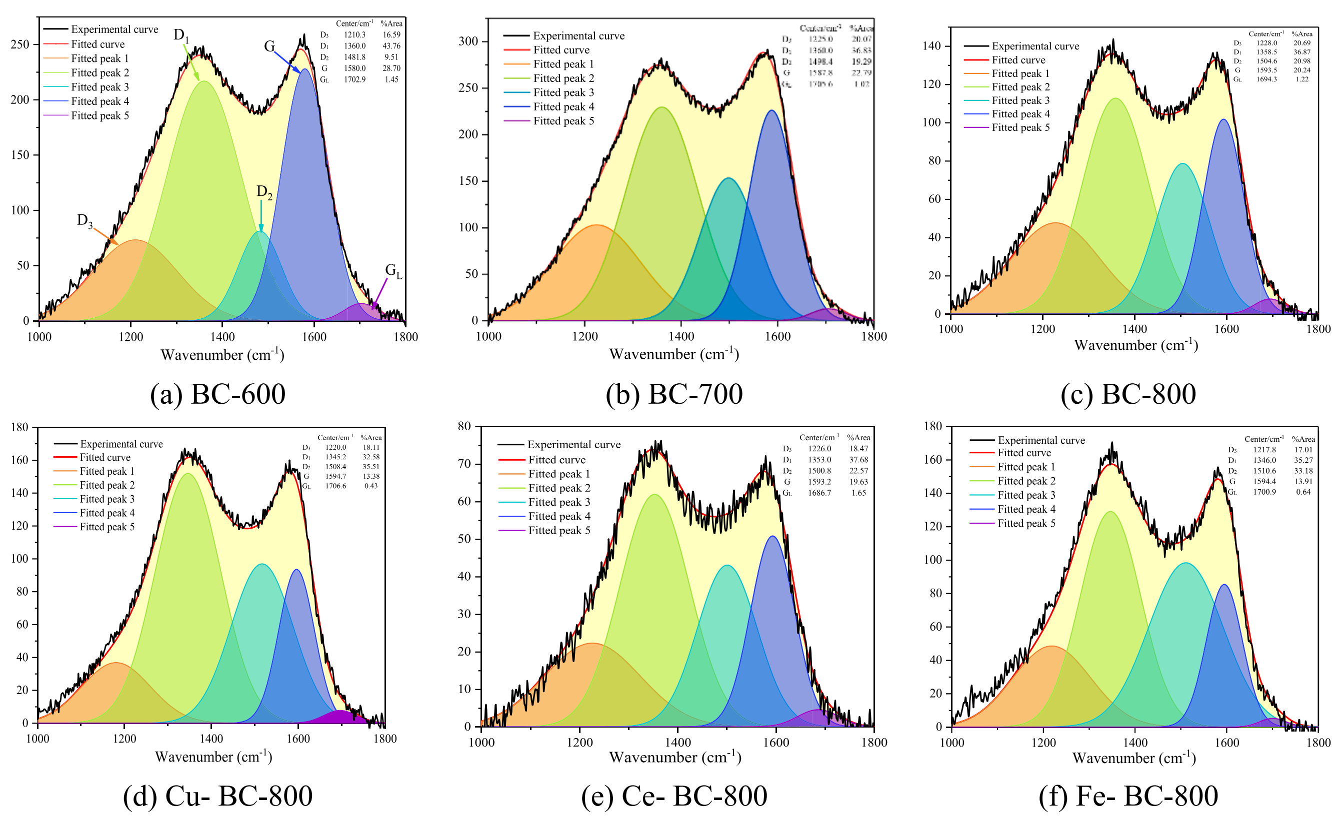

Fig. 6. Curve-fitted Raman spectra of SDBCs.

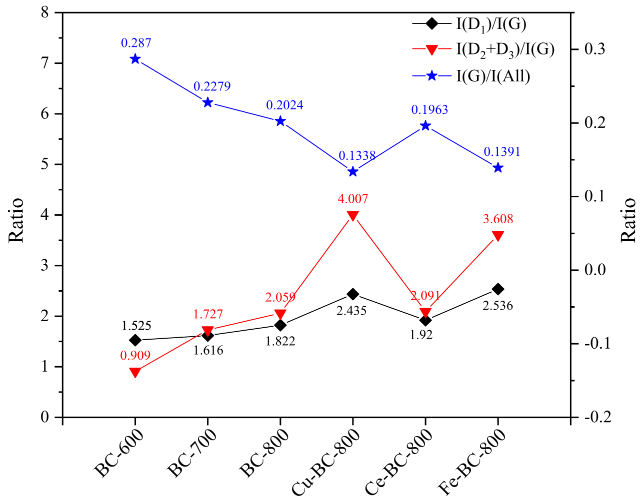

Fig. 7. Variation of $I(D_1)/I(G)$, $I(D_2 + D_3)/I(G)$ and $I(G)/I(All)$ band area ratios obtained from curve-fitting analysis of Raman spectra.

the influence of temperature on the inorganic salt species and the crystalline phase of the SDBCs was not significant. After loading different metals, the diffraction intensities of NaCl and KCl in the SDBCs were considerably reduced. The supported active metals were detected by XRD in different forms in the biochar catalyst: Cu, Fe and metal oxide CeO_2.

3.2. Catalytic pyrolysis of macroalgae over SDBCs

3.2.1. Product yields and gas compositions during catalytic pyrolysis of macroalgae

Effects of SDBCs on the yields of liquid, solid and gaseous products obtained from the *ex-situ* catalytic pyrolysis of *E. clathrate* at 550 °C are shown in Table 2. The yields of biooil, biochar and non-condensable gas obtained from the pyrolysis of *E. clathrate* were 27.22 wt%, 53.58 wt% and 18.97 wt%, respectively. The yields of biooils decreased after the SDBCs were employed in the pyrolysis process, while the yields of non-condensable gases increased. The yields of biooils with the BC-600, BC-700 and BC-800 catalysts were 26.74 wt%, 26.49 wt% and 24.22 wt%, respectively, while the yields of non-condensable gases were 20.13 wt%, 19.98 wt% and 18.98 wt%, respectively. The results demonstrated that the SDBCs could promote the gaseous products, but suppress the yields of biooils. The phenomenon was due to the catalytic cracking of the SDBCs with alkali and alkaline earth metals. Alkali and alkaline earth metals in the SDBCs, especially the K^+ and Na^+, belonged to Lewis acids, which could accelerate the thermal decomposition of carbohydrates to form low-molecular-weight products, resulting in a lower bio-oil yield [16,45]. For the Cu-BC-800, Ce-BC-800 and Fe-BC-800 catalysts, the yields of biooils were lower compared to the pyrolysis of *E. clathrate*. Biooils obtained from catalytic pyrolysis with these catalysts were selected to analyze the elements and chemical compounds, to further examine the effects of SDBCs on the liquid products.

The compositions of main gaseous products (H_2, CO, CO_2, CH_4 and C_nH_m) are shown in Table 2. The results showed that the total gas increased after using SDBCs. The main gaseous products were carbon dioxide, carbon monoxide and hydrogen in all cases. SDBCs could promote the release of CO_2, which might be ascribed to the deoxygenation of biomass pyrolysis volatiles promoted by biochar catalysts.

3.2.2. Chemical compound distribution of biooils during catalytic pyrolysis of macroalgae

Element analysis of biooils are shown in Table 3. The biooil obtained from the direct pyrolysis of *E. clathrate* comprised low carbon (39.01 wt %) and high oxygen content (35.41 wt%). After using the SDBCs, carbon contents in biooils showed a significant increase, which reached a peak of 59.56 wt% in the case of BC-600 catalysts. Oxygen contents decreased after using the biochar catalysts, especially in the case of Cu-BC-800 catalyst. In addition, the nitrogen contents of biooils obtained from catalytic pyrolysis over SDBCs decreased in comparison to the control case. The $(H/C)_{eff}$ in the control was only 0.167, which increased after using SDBCs. The highest $(H/C)_{eff}$ recorded 0.503 when Cu-BC-800 was used. The high heating values of biooils were also promoted after using the SDBCs. Therefore, all of the SDBCs could promote both $(H/C)_{eff}$ and HHVs, which was resulted from removing the oxygen content and increasing carbon contents of biooils.

GC–MS analysis was conducted to detect the chemical content of *E. clathrate* catalytic pyrolysis biooils. The detailed compounds were listed in Table S1 from the supplemental material. The compounds could be classified into the following groups; carboxylic acids, aldehydes, aliphatic hydrocarbons, sugars, aromatic hydrocarbons, esters, ketones, N-containing, phenols and others. The relative contents of those groups are presented in Table 4. The main components of biooil from *E. clathrate* direct pyrolysis were carboxylic acids, N-containing compounds and phenols, with the relative contents of 32.54%, 28.67% and

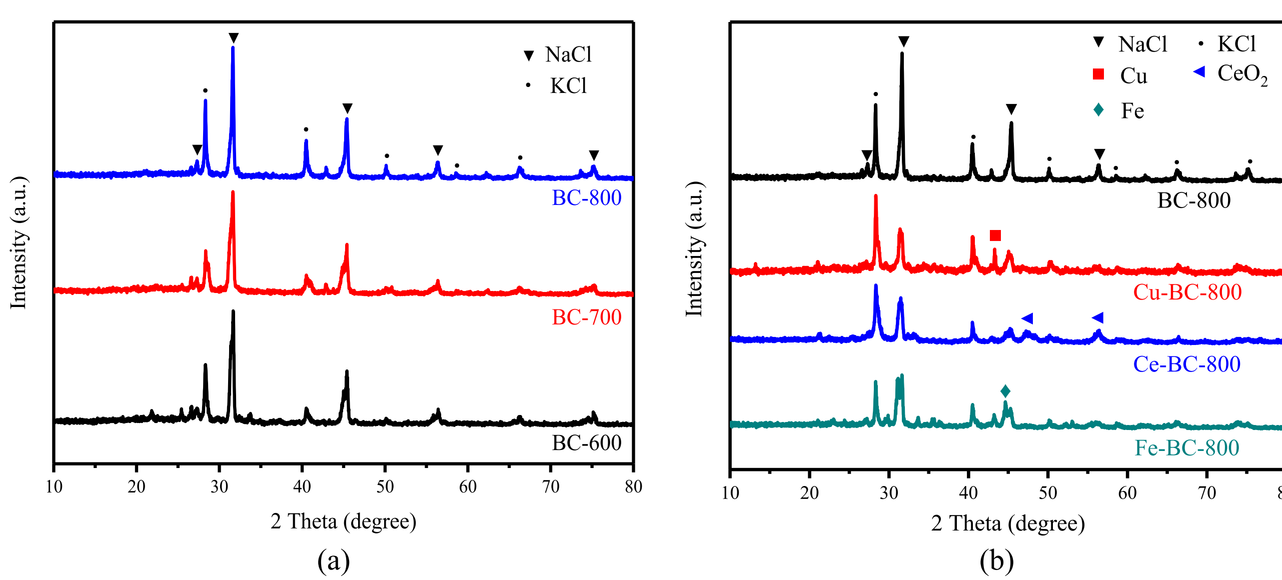

Fig. 8. XRD patterns of of (a) BC-600, BC-700, BC-800; (b) Cu-BC-800, Ce-BC-800, Fe-BC-800.

Table 2
Effects of catalysts on the yields of the product yields and gas compositions.

	Control	BC-600	BC-700	BC-800	Cu-BC-800	Ce-BC-800	Fe-BC-800
Product yields (wt.%)[a]							
Bio-oil	27.22	26.74	26.49	24.22	25.45	23.54	23.58
Bio-char	53.58	53.22	53.69	53.39	53.10	52.92	52.88
Non-condensable gas	18.97	20.13	19.98	18.98	20.88	20.28	20.43
Coke	–	2.66	2.53	2.55	2.84	3.69	3.59
Total yields	99.77	102.76	102.68	99.14	102.27	100.43	100.49
Gas compositions (mmol/g E. clathrate)							
H_2	0.830	1.480	1.130	1.036	1.932	1.354	1.957
CO	2.162	1.007	2.523	2.174	2.709	1.752	2.479
CO_2	2.468	3.409	2.477	2.483	2.465	3.055	2.506
CH_4	0.473	0.513	0.476	0.467	0.528	0.436	0.537
C_nH_m[b]	0.335	0.352	0.306	0.301	0.362	0.284	0.362
Total gas	6.268	6.761	6.912	6.460	7.996	6.882	7.841

[a] All the average values are within the standard deviation of ±3.0%.
[b] $C_nH_m = C_2H_6 + C_2H_4 + C_3H_8 + C_3H_6$.

Table 3
Characteristics of macroalgae and obtained biooils over SDBCs.

Elements (wt.% dry basis)[a]	Biooils						
	Control	BC-600	BC-700	BC-800	Cu- BC-800	Ce- BC-800	Fe- BC-800
C	39.01	59.56	58.68	57.49	59.47	58.06	55.58
H	8.46	7.40	7.44	7.50	7.60	7.56	7.33
N	15.95	10.25	10.67	9.94	11.09	11.09	11.93
S	1.17	0.00	0.00	0.00	0.00	0.00	0.00
O[b]	35.41	22.79	23.21	25.07	21.84	23.29	25.16
$(H/C)_{eff}$	0.167	0.474	0.461	0.467	0.503	0.470	0.352
HHV/(MJ/kg)	21.660	28.398	28.118	27.560	28.723	28.051	26.794

[a] Weight %.
[b] O = 100-C–H-N-S-Ash.

9.22%, respectively. The main compound of carboxylic acids in biooil from *E. clathrate* was n-hexadecanoic acid, whose relative content was 30.17% (Table S1). After using the SDBCs, sugars, esters and N-containing compounds were observed as the main groups. Interestingly, there were very low relative contents of carboxylic acids detected in biooils. The relative content of carboxylic acids was decreased to 2.11% after using BC-600, which consisted of tetradecanoic acid (1.22%) and 2,3,4-trimethylpentanoic acid (0.89%). Also, the relative contents of carboxylic acids were decreased to zero when BC-700, BC-800, Cu-BC-800, Ce-BC-800 and Fe-BC-800 were used. Therefore, the oxygen contents in biooils after using the SDBCs decreased, which was consistent with the results from element analysis. This indicated that the SDBCs had excellent performance for suppressing the formation of carboxylic acids, especially n-hexadecanoic acid, in biooils. The same phenomenon of removal long-chain fatty acids in biooil was reported by Chen W et al. In their experiments, the biochar catalyst from the fast pyrolysis of bamboo wastes employed to remove long-chain fatty acids. The relative content of fatty acids was decreased from 59% to 30%. The long-chain fatty acids intermediates (the palmitoleic acid and oleic acid) with carbon–carbon double bond had higher activity, thus the biochar catalyst could promote the dehydration and cyclization of those intermediates. However, there was still 30% of long-chain fatty acids after using biochar catalyst. The reason was that the long-chain fatty acids intermediates with alkyl-chain (n-hexadecanoic acid and tetradecanoic acid) had higher stability, and did not react with biochar catalyst completely. Except the similarity, our results had a different point, which exhibited that the stable long-chain fatty acids intermediates with alkyl-chain (n-hexadecanoic acid and tetradecanoic acid) also react with the –OH groups abundant on the surface of the SDBCs from FTIR analysis through bond cleavage and esterification reactions, forming the aliphatic hydrocarbons and the long-chain fatty acid methyl esters (FAMEs). For the different results, the reasons were proposed to be the different chemical properties of the biochar catalysts from the fast pyrolysis of bamboo wastes and seaweed, which consisted of different contents of oxygen-containing functional groups (–OH) and inorganic minerals.

Compared with the relative content of esters in the control case (2.22%), the esters were increased sharply by the SDBCs, which ranged

Table 4
The relative contents of different groups in biooils of macroalgae over the biochar catalysts.

Groups	Relative content (%)						
	Control	BC-600	BC-700	BC-800	Cu-BC-800	Ce-BC-800	Fe-BC-800
Carboxylic acids	32.54	2.11	0	0	0	0	0
Aldehydes/ Ketones	11.79	1.01	2.39	4.74	0	0	8.85
Aliphatic hydrocarbons	6.45	7.75	4.83	11.12	9.64	6.48	6.47
Aromatic hydrocarbons	3.78	0	0	0	3.54	0	0
Esters	2.22	8.93	33.60	20.17	35.77	33.55	32.49
Sugars	4.12	13.36	24.06	20.53	17.32	16.75	0
N-containing	28.67	45.12	28.69	33.86	30.94	38.27	44.78
Phenols	9.22	18.23	6.43	9.58	0	4.94	7.42
Others	1.21	3.5	0	0	2.80	0	0

from 8.93% to 35.77%. The effect of BC-600 on promoting esters was less than the other SDBCs. BC-600 showed the selectivity for heptadecanoic acid methyl ester, whose content was 5.57%, while BC-700 and BC-800 showed excellent performances for tridecanoic acid methyl ester, whose content were 13.98% and 8.90%, respectively. Cu, Ce and Fe loaded in the BC-800 catalyst showed a synergistic effect with amorphous carbon support containing oxygen-containing functional groups and inorganic minerals on the production of esters. The relative content of esters in the case of BC-800 catalyst was 20.17%, which increased to 35.77%, 33.55% and 32.49%, after the metals Cu, Ce and Fe, respectively, were employed to modify BC-800. Cu-BC-800 exhibited excellent performance for tridecanoic acid methyl ester, whose content was 16.49%. Additionally, Ce-BC-800 and Fe-BC-800 showed excellent performance for decanoic acid methyl ester, whose contents were 13.35% and 10.97%, respectively (Table S1).

Compared with the relative content of sugars in biooil obtained from the direct pyrolysis of *E. clathrate*, sugars significantly increased after using the SDBCs except for the case of Fe-BC-800. The relative content of sugars in the control was 4.12%, while the contents of sugars after using BC-600, BC-700, BC-800, Cu-BC-800, and Ce-BC-800 ranged from 13.36% to 24.06%. The metals Cu, Ce and Fe loaded on the BC-800 support seemed to suppress the effect of SDBCs for the production of sugars. The Fe-BC-800 catalyst showed the worse inhibition of sugar production, which was not detected in biooil. For the formation of methyl-2,4-di-O-methylαd-glucopyranoside, BC-600, BC-700, BC-800, Cu-BC-800, and Ce-BC-800 catalysts showed significant selectivity from Table S1. Alkali and alkaline earth metal ions, including K^+, Na^+, Mg^{2+} and Ca^{2+} existed in the SDBCs, played the role of Lewis acid sites in the pyrolysis process, which could enhance dehydration for production of sugar compounds [16]. In addition, the aldehydes, especially the 5-methyl-furfural (5-MFF) and furfural (FF), were also decreased by the SDBCs. The reason was that the alkali and alkaline earth metal ions in the SDBCs influenced the pyrolysis process of carbohydrates in the macroalgae, which inhibited the formation of 5-MFF and FF. A similar phenomenon was observed in a previous research [46], which pointed out that the formation of levoglucosan during cellulose pyrolysis were inhibited by K^+, Na^+, Mg^{2+} and Ca^{2+}.

The effects of Cu, Ce and Fe loaded on BC-800 catalyst on the chemical compound distributions were concluded from Table 4. Compared with the chemical compound distributions of aliphatic hydrocarbons and sugars, the metals Cu, Ce and Fe had an inhibition effect followed the order of: Fe > Ce > Cu. The relative contents of aliphatic hydrocarbons and sugars when Cu-BC-800 was used showed a gentle decrease of 1.48% and 3.21%, respectively. Also, the metals Cu, Ce and Fe had an inhibition effect of forming phenolic compounds followed the order of: Cu > Ce > Fe. The relative content of phenols was decreased to zero after the Cu-BC-800 catalyst employed, so as the aldehydes and ketones. The N-containing compounds decreased gently after the Cu-BC-800 catalyst was used, while increased in the cases of the Ce-BC-800 and Fe-BC-800 catalysts. On the other hand, the Cu-BC-800 catalyst could increase the relative content of esters in biooils in the following order: Cu > Ce > Fe. Among all the SDBCs, the relative content of esters reached a peak of 35.77% in the case of the Cu-BC-800 catalyst, simultaneously, the oxygen content was the lowest, and the $(H/C)_{eff}$ and HHV were the highest in all conditions. The results indicated that the metal Cu was more suitable to modify seaweed-derived biochar for the ester/sugar-abundant biooils than Ce or Fe.

Therefore, the SDBCs could decrease the carboxylic acids significantly and increase the sugars and esters to a different extent in biooils, which was mainly due to the alkali and alkaline earth metals and oxygen-containing functional groups on the surface of biochar catalysts. In addition, the metals Cu, Ce and Fe loaded on the BC-800 catalyst showed different effects on the chemical compound distributions of biooils.

3.2.3. Chemical reaction paths during catalytic pyrolysis of macroalgae

The possible reaction pathways of macroalgae volatiles over different SDBCs were proposed as shown in Fig. 9. The most apparent change of biooil compounds was the decrease of carboxylic acids. It was reported that lipid was the chief source for producing long-chain fatty acids [47]. The decrement of carboxylic acids (especially n-hexadecanoic acid) could be related to the increment of esters. From the perspective of structural similarity of long-chain fatty acids and FAMEs, a significant amount of hydroxyl groups existed in biochar could react with carboxylic group to form FAMEs. Another possible conversion pathway for carboxylic acid should be decarboxylation reaction [33,48]. As shown in GC–MS results, most linear hydrocarbons should be converted from carboxylic acids. In addition, the formation of neophytadiene should be a typical product of chlorophyll pyrolysis [36]. The relative content of aliphatic hydrocarbons after the Cu/Ce/Fe-BC-800 catalysts were used was lower than that over BC-800, illustrating that metallic active sites Cu, CeO_2 and Fe could accelerate C–C bond scission. As one of the important constituents in seaweed, polysaccharides could generate a lot of furan derivatives through dehydration and ring opening reactions, such as aldol reaction [49]. The formation of 5-MFF could be attribute to the decomposition of polysaccharides and the dehydration and isomerization of monocarbohydrates. The FF could also be converted from polysaccharides in macroalgae. Whereas, furan derivatives could not be found in catalytic products, indicating that the isomerization including ring opening reaction or enol-keto tautomerization were inhibited. Some sugars were detected in catalytic pyrolyzed biooils. Amongst, the most abundant compound was the methyl α-D-glucoside, whose relative content ranged from 12.03% to 24.06%, while its content was zero in non-catalytic pyrolysis which is contrasting to catalytic condition. This could be owing to the α-carbon in the sugar initially reacted with active groups in the seaweed-derived biochar, such as hydroxyl group, which could prevent aldol reaction °Ccurrence and the formation of furfurals [50]. The metallic compositions loaded on the BC-800, namely, Cu, CeO_2 and Fe, had similar effects on promoting the formation of esters and suppressing the formation of aliphatic hydrocarbons and phenols. Besides, those metallic compositions had different effects on the reaction pathways and products distribution of biooils during catalytic pyrolysis. Both Cu and CeO_2 had strong inhibition effects on the aldehydes and ketones, which might be owing to the reaction between -C = O- units in ketones/aldehydes and those active sites. In addition, Cu loaded on the BC-800 could promote the formation of aromatic hydrocarbons in biooil, while the promotion was not operative after using CeO_2 and Fe as the active compositions. In general, the loaded metallic compositions had particular effects on the C–C bond and -C = O- units in the pyrolysis volatiles. We suggested that the Cu-BC-800 catalyst was a promising metal modified biochar catalyst for producing acid-free biooils containing abundant esters and sugars.

4. Conclusions

A series of environmentally friendly, easily obtainable and less cost SDBCs were prepared at three different temperatures of 600, 700 and 800 °C for upgrading the pyrolysis biooil from macroalgae. Multiple characterizations of the biochar catalysts indicated that Cu, Ce and Fe modified biochar catalysts at the temperature of 800 °C displayed highly developed porous structures and more aromatic rings with no less than 6 benzene rings along with defects and heteroatoms. The most inorganic minerals were mostly present in the form of NaCl and KCl in all the SDBCs through XRD analysis. Subsequently, the performance of all the SDBCs was examined through ex-situ catalytic pyrolysis of *Enteromorpha clathrate*. The results demonstrated that the catalysts could promote the gaseous products, but suppress the yields of biooils due to the existence of NaCl and KCl. The biooils from catalytic pyrolysis exhibited higher $(H/C)_{eff}$ and HHVs than the control. The chemical compound distributions of biooils were also affected significantly, containing abundant

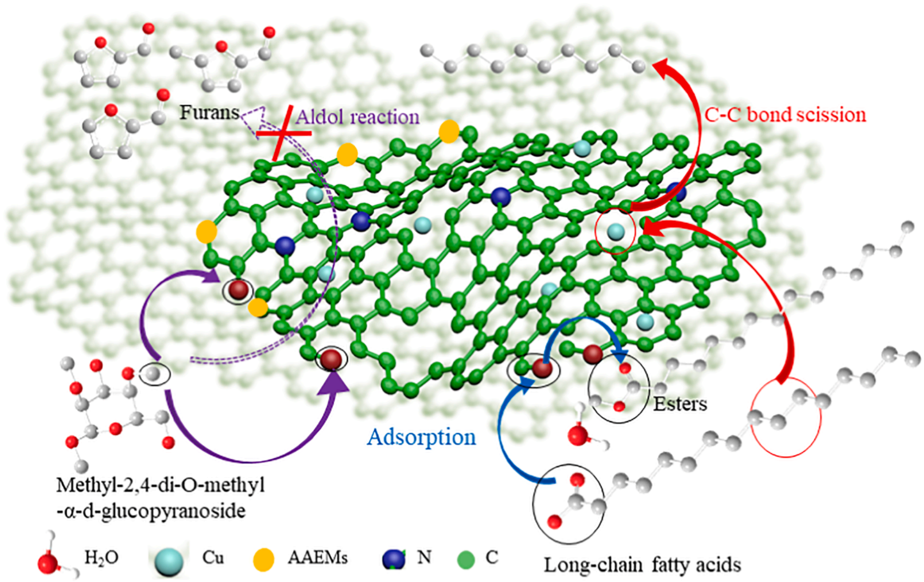

Fig. 9. Mechanism scheme of macroalgae volatiles over different SDBCs.

esters and sugars, as high as 35.77% and 17.32%, respectively in the case of the Cu-BC-800 catalyst. The carboxylic acids were decreased significantly after using the SDBCs. The mechanism is that a significant amount of hydroxyl groups existed in biochar could react with carboxylic group to form FAMEs. Additionally, the metals Cu, Ce and Fe loaded on the BC-800 catalyst the loaded metallic compositions had particular effects on the C–C bond and - C = O - units in the pyrolysis volatiles. We suggested that the Cu-BC-800 catalyst was a promising metal modified biochar catalyst for producing acid-free biooils containing abundant esters and sugars.

CRediT authorship contribution statement

Bin Cao: Conceptualization, Data curation, Writing - original draft. **Jianping Yuan:** Investigation, Supervision. **Ding Jiang:** Writing - original draft. **Shuang Wang:** Conceptualization, Methodology, Supervision. **Bahram Barati:** Writing - review & editing. **Yamin Hu:** Writing - review & editing. **Chuan Yuan:** Data curation, Supervision. **Xun Gong:** Methodology, Supervision. **Qian Wang:** Methodology, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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